

Whitelist Manager

Quick Demo Guide

Evaluate the app in Docker before production deployment

One command to go from zero to a working demo

Security Engineering Team

1. Prerequisites

Before running the demo, ensure the following are installed and running:

- Docker Desktop (Windows, macOS, or Linux)
- Bash shell (Git Bash on Windows, or native terminal on Linux/macOS)
- Approximately 1 GB of free disk space for the Splunk Docker image
- Internet connection (to pull the Splunk image on first run)

Note: The demo uses ports 9000 (Web UI) and 9089 (API). Make sure these ports are not in use by other applications. The demo container is completely separate from any development environment.

2. Starting the Demo

From the root of the wl_manager repository, run:

```
bash demo/demo.sh
```

The script will automatically:

1. Verify Docker is running
2. Build the .spl package (if not already built)
3. Start a Splunk 9.3.1 container
4. Install the Whitelist Manager app from the .spl
5. Create demo users with different roles (editor, viewer, admin)
6. Seed three demo detection rules with sample whitelist data
7. Print the login URL and credentials

The entire process takes approximately 2-3 minutes (longer on first run while Docker pulls the Splunk image).

Login Credentials

URL: <http://localhost:9000>

Username	Password	Role	Can Do
admin	Chang3d!	Splunk admin	Everything
analyst1	Chang3d!	wl_analyst_editor	Edit whitelists
viewer1	Chang3d!	wl_analyst_viewer	View only

wladmin1	Chang3d!	wl_admin	Approve/reject, config
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3. What to Try

3.1 Browse Whitelists

1. Log in to Splunk Web at <http://localhost:9000>
2. Navigate to Apps > Whitelist Manager
3. Click the "Detection Rule" dropdown and select Brute_Force_Login
4. The CSV File dropdown auto-populates. Select brute_force_whitelist.csv
5. The table loads with 4-5 sample rows including expiration dates

Tip: Notice the yellow banner at the top indicating that one expired row was automatically removed on load. This demonstrates the auto-expiration feature.

3.2 Edit a Cell

1. Click on any cell in the table (e.g., a Comment field)
2. Type a new value
3. Click "Save Changes"
4. Enter a comment explaining the change (e.g., "Demo edit")
5. Review the Git-style diff summary that appears below the table

3.3 Add a Row

1. Click "+ Add Row"
2. Fill in the fields in the new empty row
3. Click "Save Changes" and provide a comment

3.4 Remove a Row

1. Click "Remove" on any row's Actions column
2. Enter a reason when prompted (required for audit)
3. Notice the 10-second Undo bar that appears

3.5 Try the Date Picker

1. Select the Brute_Force_Login or Impossible_Travel rule
2. Click on any cell in the Expires column

3. A date/time picker appears with preset buttons (7 Days, 30 Days, etc.)
4. Click "30 Days" to set an expiration 30 days from now
5. Or pick a manual date/time and click "Apply"

3.6 Version Control and Revert

1. After making a few changes, look at the Revert to Version dropdown
2. It shows timestamped snapshots of previous versions
3. Select a previous version to see what would change
4. Enter a reason and click Revert to restore that version

3.7 Test Read-Only Access (as viewer1)

1. Log out and log in as viewer1
2. Navigate to Whitelist Manager - you can view but not edit
3. The Save, Add Row, and Remove buttons are disabled
4. The "Control Panel" tab is not visible in the navigation

4. Approval Workflow Demo

Whitelist Manager enforces an approval workflow for high-impact changes. Bulk operations above configurable thresholds require admin approval.

4.1 Trigger an Approval Gate (as analyst1)

1. Log in as analyst1
2. Go to Whitelist Manager and select Suspicious_Process
3. Select all rows using the header checkbox
4. Click "Remove Selected (4)"
5. Enter a reason when prompted (e.g., "Cleanup old entries")
6. Since 4 rows exceeds the threshold (3), you will see: "Removing 4 rows requires admin approval"
7. Click "Submit for Approval" and add a justification
8. The CSV is now locked until an admin reviews the request

4.2 Approve the Request (as wladmin1)

1. Log out and log in as wladmin1
2. Click the "Control Panel" tab in the navigation bar
3. The Approval Queue shows the pending request from analyst1
4. Click "Approve" (or "Reject" with a reason)

5. The approved operation executes automatically
6. analyst1 receives a notification about the decision

Note: Self-approval is prevented. If analyst1 also has admin roles, they still cannot approve their own request. Another admin must review it.

4.3 Check Daily Limits

1. In the Control Panel, click the "Analyst Usage" tab
2. See analyst1's usage counters for today
3. Click "Limits & Permissions" to see and adjust thresholds
4. Try changing the bulk removal threshold from 3 to 5

5. Viewing the Audit Trail

1. Click the "Audit Trail" tab in the app navigation bar
2. The dashboard shows all changes made during your demo session
3. Use the filters at the top to narrow by analyst, rule, or action type
4. The Summary Statistics show total changes, adds, removes, and edits
5. Check the "Expiring Soon" panel to see rows approaching their expiration date

You can also search audit events directly in Splunk's Search & Reporting app:

```
index=wl_audit sourcetype=wl_audit  
| table timestamp analyst action detection_rule csv_file comment
```

6. Demo Data Reference

The demo seeds three detection rules with sample whitelists:

Brute_Force_Login

CSV: brute_force_whitelist.csv

Columns: user, src_ip, threshold, Comment, Expires

Features demonstrated: expiration dates, auto-removal of expired rows, date picker

Suspicious_Process

CSV: suspicious_process_whitelist.csv

Columns: host, process_name, user, Comment

Features demonstrated: simple whitelist without expiration, add/edit/remove rows

Impossible_Travel

CSV: impossible_travel_whitelist.csv

Columns: user, src_country, Comment, Expires

Features demonstrated: expiration dates, date picker presets

7. Stopping the Demo

Stop and Remove Container

This stops the container and removes it, but keeps the data volume for next time:

```
bash demo/demo.sh --stop
```

Full Cleanup

This removes both the container and the data volume (complete reset):

```
bash demo/demo.sh --clean
```

Restart the Demo

After stopping, run the demo again to get a fresh environment:

```
bash demo/demo.sh
```

If you used --stop (not --clean), the demo restarts faster because the Splunk image is already cached and the volume retains configuration.

8. Troubleshooting

Docker is not running

If you see "[ERROR] Docker is not running", start Docker Desktop and wait for it to fully initialize before running the script again.

Port already in use

If port 9000 or 9089 is already in use, stop the conflicting service or edit the WEB_PORT / API_PORT variables at the top of demo.sh.

Splunk did not start in time

On slower machines or first run (while pulling the image), Splunk may take longer to start. Run the script again - it will detect the existing container and retry. You can also check logs with:

```
docker logs wl_manager_demo
```

No detection rules in dropdown

If the dropdown is empty after login, Splunk may still be loading. Wait 30 seconds and refresh the page. If it persists, restart the demo container:

```
docker restart wl_manager_demo
```